



SensAIR: Transforming Pediatric Respiratory Care with Continuous, Objective Distress Detection

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RICE



Texas Children's Hospital

Monitoring Challenges

- Respiratory distress can lead to sudden deterioration that escalates into **life-threatening** conditions

24.2 million pediatric ED visits/year¹

30% with respiratory issues

7.26 million pediatric ED visits related to respiratory issues/year in the US

- Standard of Care (SOC):** manual monitoring of respiratory rate (RR) and work of breathing (WoB) to identify respiratory decline
- Labor-intensive** and **intermittent** process, places a significant burden on nursing staff and delays detection of patient decline

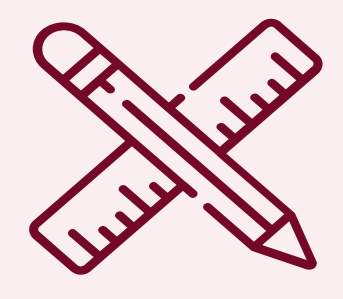
Mission Statement

Develop a device to **continuously monitor** and **quantify** work of breathing in pediatric patients to **monitor for respiratory decline** and **treatment response**

Design Specifications



Non-Invasive
0 entries into body



Accurate Quantification
+/- 2 breaths/min of SOC



Wireless
0 physical tethering



Comfortable
<100 grams & fits 5th - 95th percentile

Clinical Rationale for Multi-Sensors

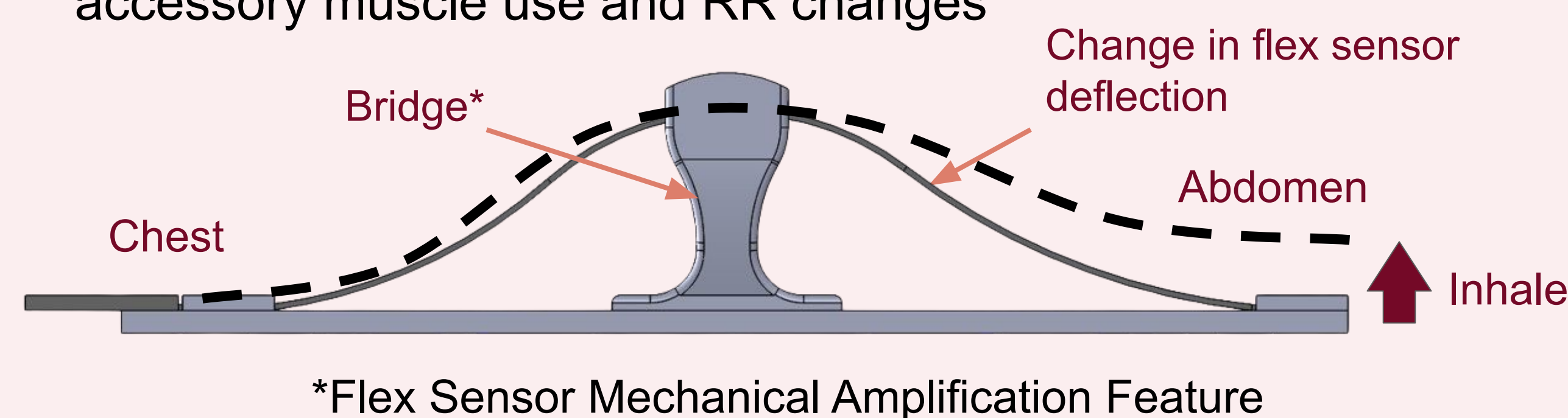
- Components of WoB include:
 - Changes in respiratory rate
 - Accessory muscle use
- Rising CO₂ (hypercapnia) is an earlier sign of impending respiratory failure than hypoxia (declining SpO₂)

CO₂ sensor:

- CO₂ is a direct indicator of ventilation

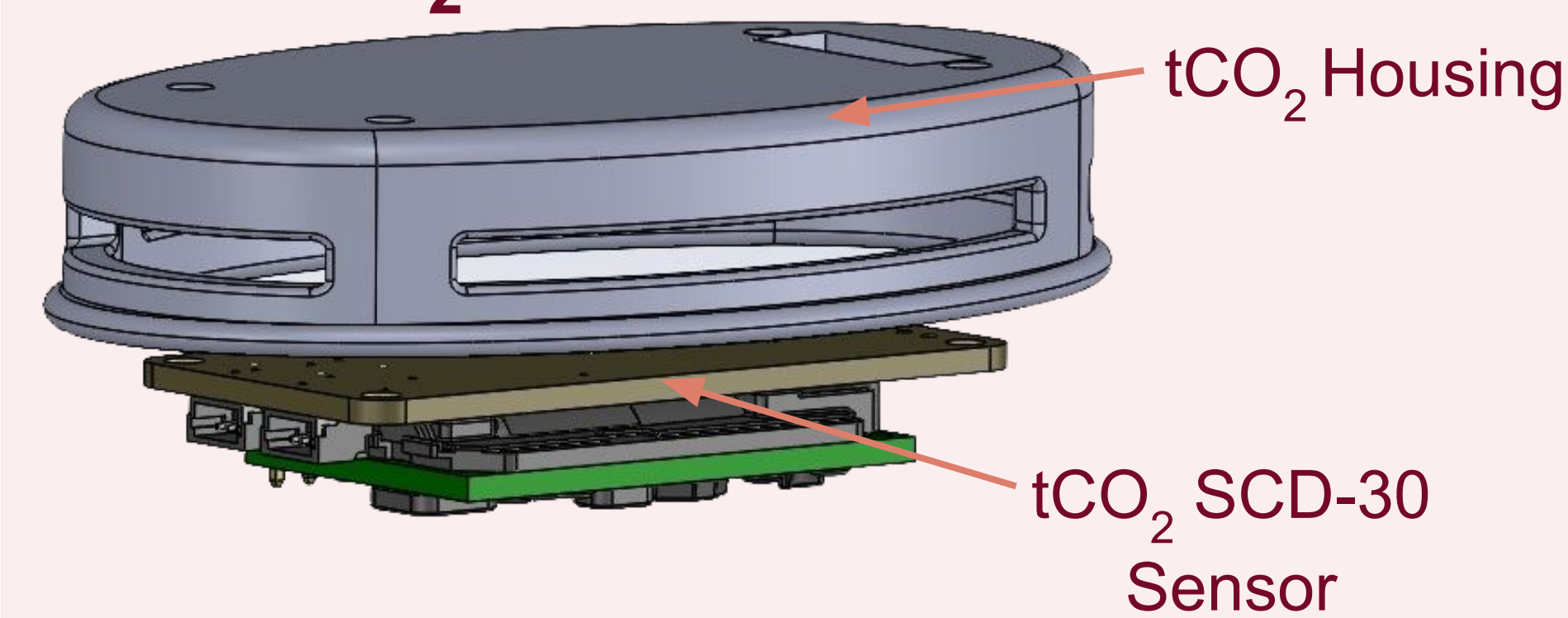
Flex Sensor:

- Measures **thoracoabdominal dissociation** quantifying accessory muscle use and RR changes

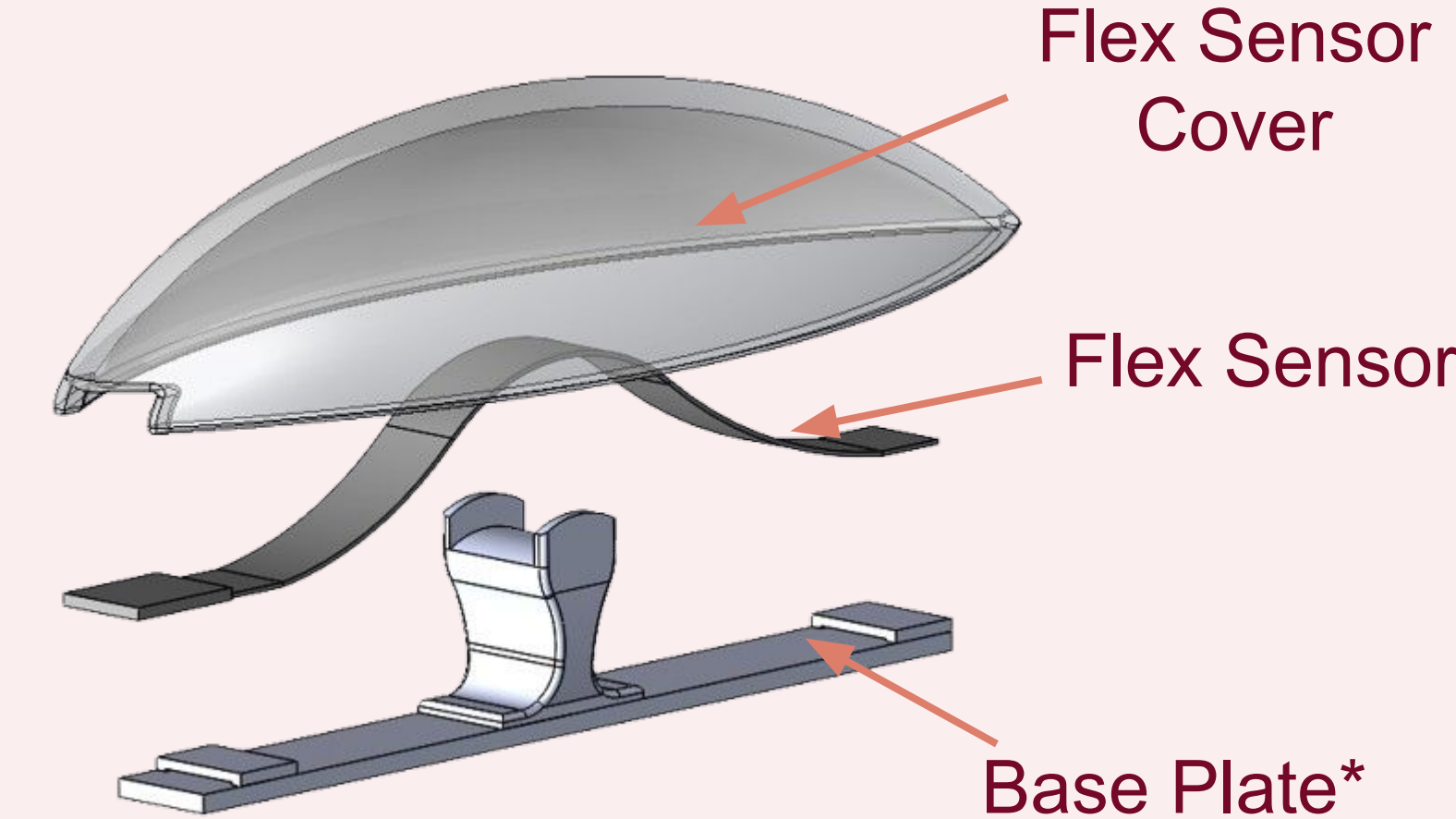


Our Solution: SensAir

A. transcutaneous CO₂ (tCO₂) Sensor

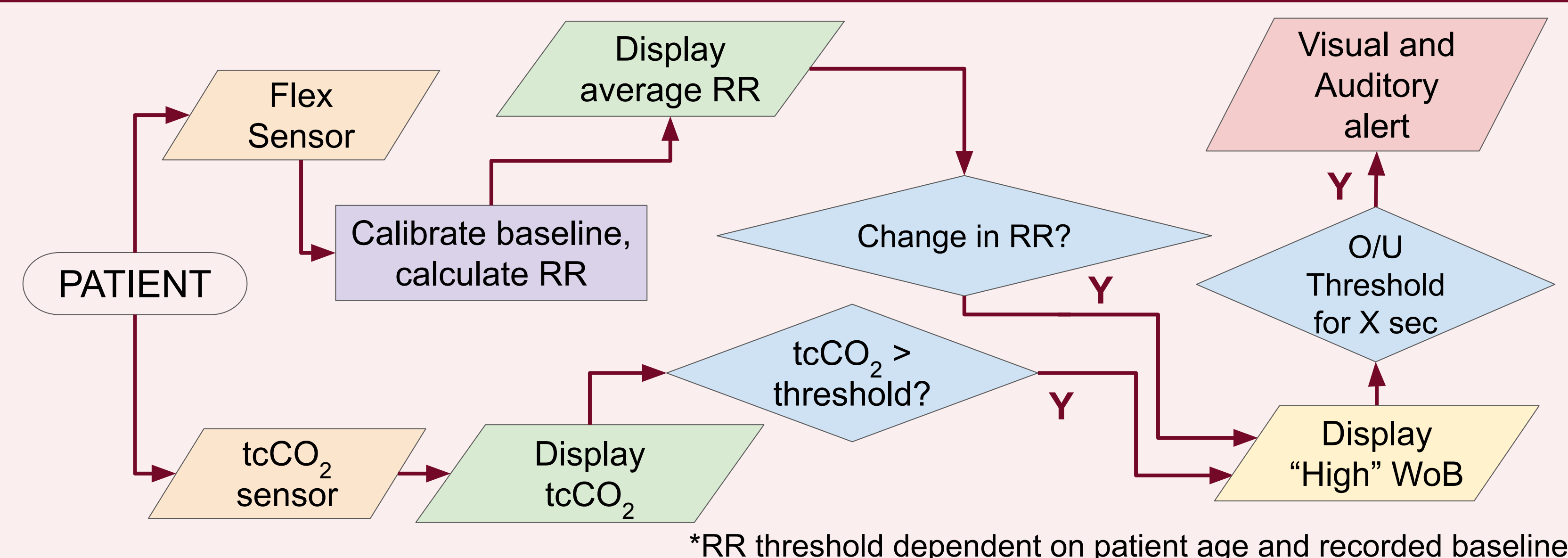


B. Flex Sensor



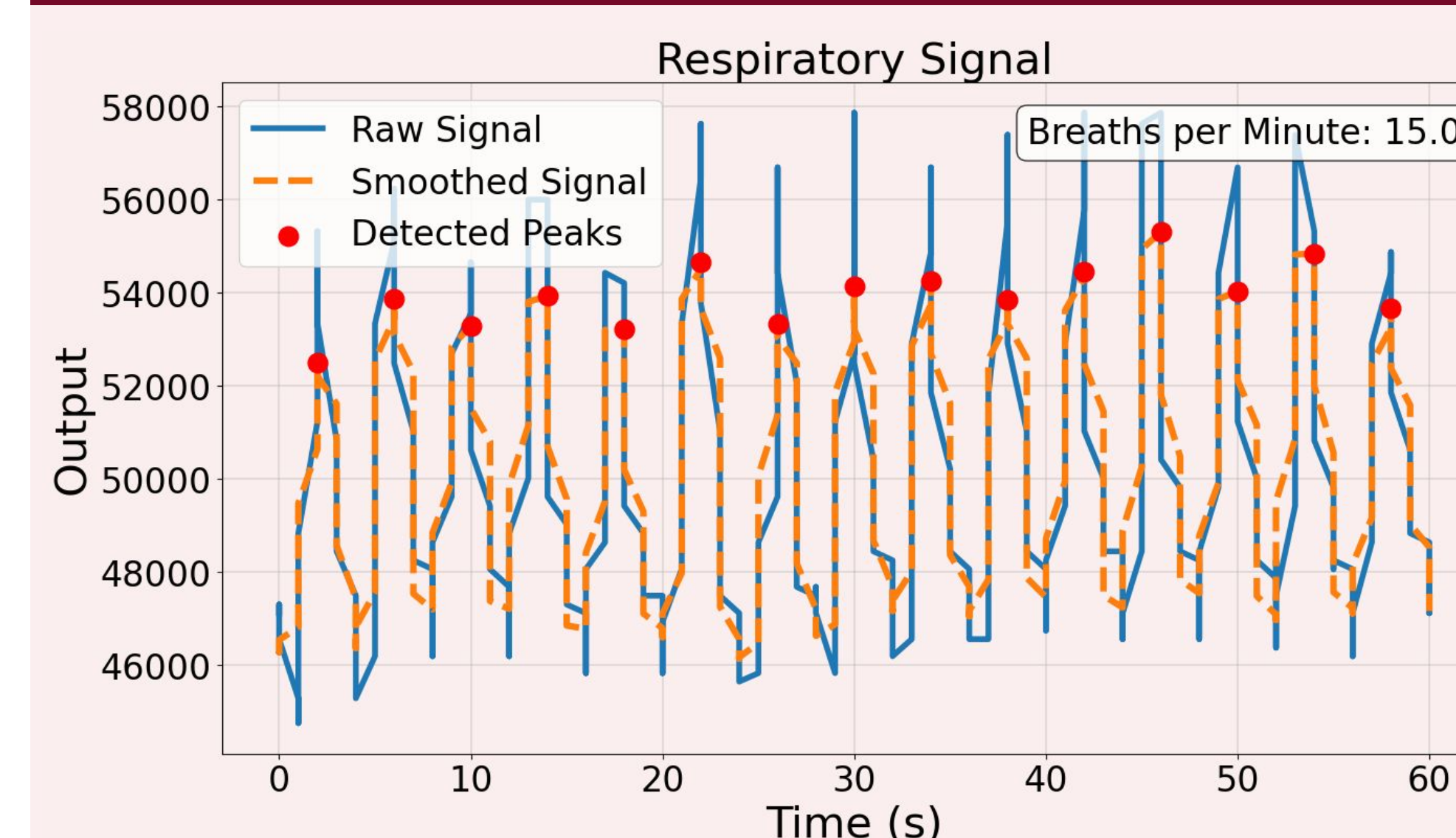
*Amplify subtle diaphragm movements using a pre-loaded bridge configuration

Data Algorithm Flow Chart



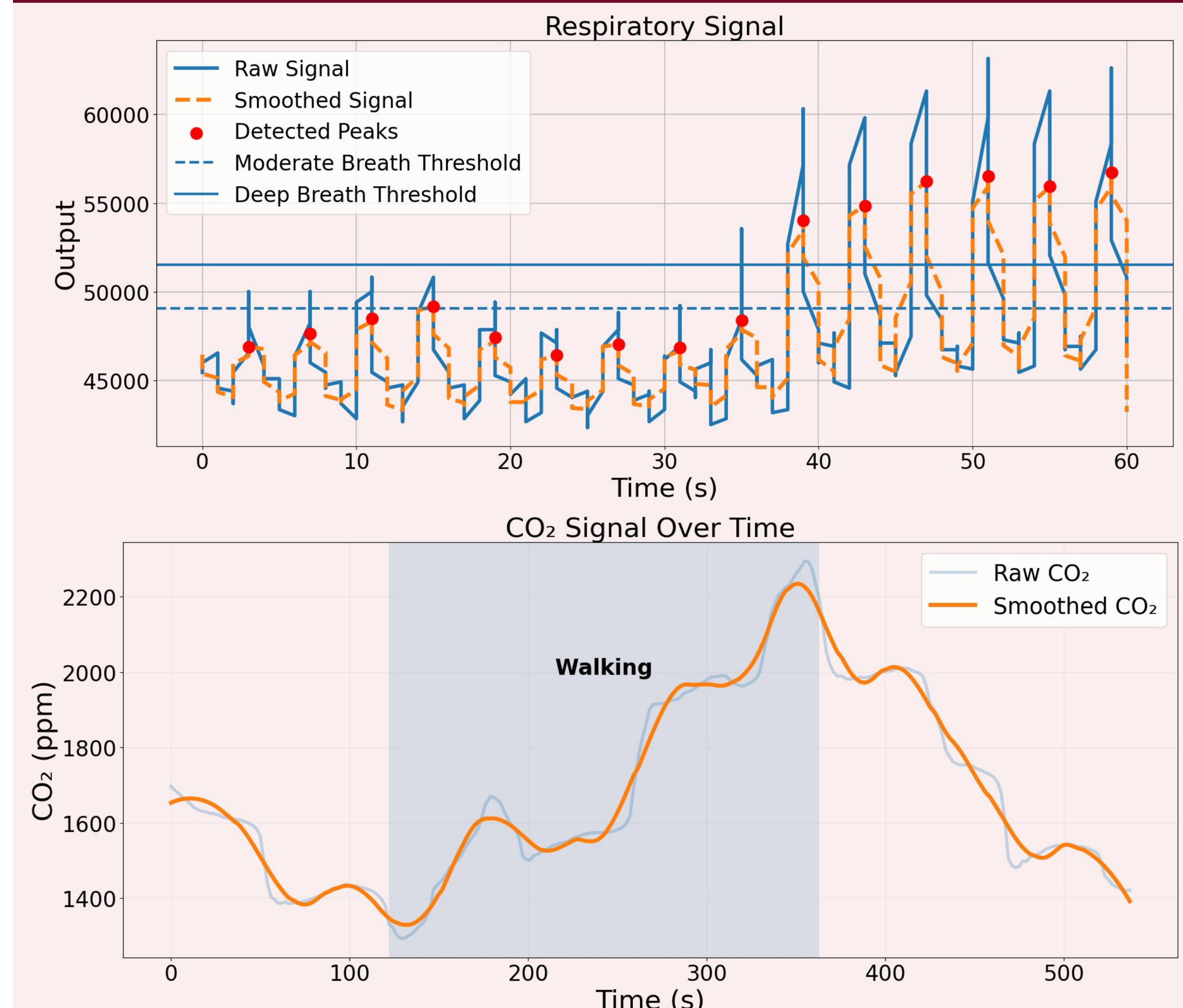
*RR threshold dependent on patient age and recorded baseline

Signal Processing



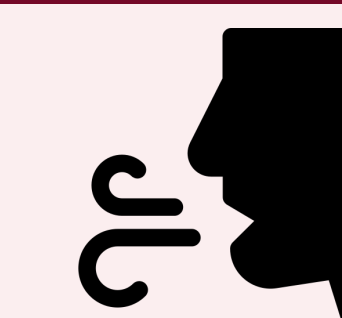
- Signal Processing: Implemented a smoothing algorithm to filter noise and identify peak detection
- Different positions have resulted in different muscle accessory use²
 - Amplitude will vary depending on seated or standing

Testing Results

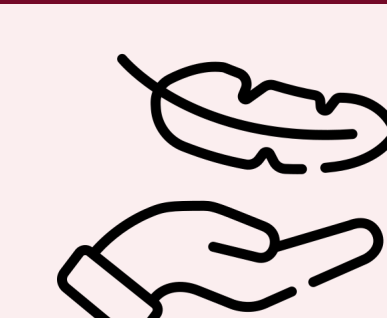


- Clear identification of breathing intensity
- Significant increase in CO₂ during physical activity

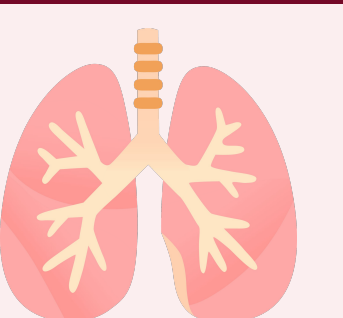
Device Summary



Quantifies **WoB** levels

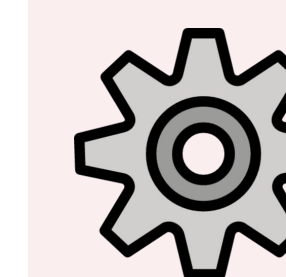


Total Weight: **80g**

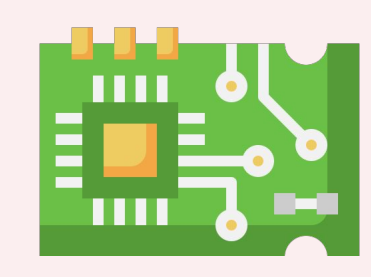


97.5% peak detection of RR

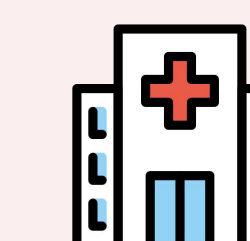
Next Steps



Refine and miniaturize design into a wireless wearable



Integrate both sensors into a single housing



Launch clinical trials with user interface validation and usability assessment

References

- [1] Centers for Disease Control and Prevention. (2021). National Hospital Ambulatory Medical Care Survey: 2021 emergency department summary tables.
- [2] Ackermann BJ, O'Dwyer N, Halaki M. doi: 10.3389/fpsyg.2014.00913.

Acknowledgements

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